

Maryam Torkashvand

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PROFESSIONAL SUMMARY

- GIS / spatial modeling specialist with 4 years of experience building repeatable geospatial workflows across vector, raster, and network datasets.
- Strong Python and SQL (PostgreSQL/PostGIS) for geospatial automation and large-scale data management.
- Proficient in using ArcGIS tools to produce maps and visualizations for technical and non-technical audiences.
- Contributed to 5+ interdisciplinary projects by creating workflows for geospatial data engineering, spatial analysis, and visualization to support research in GIS, hydrology, and network science.

PROFESSIONAL EXPERIENCE

Graduate Research Assistant (Spatial Data Specialist)

School of Earth, Environment, and Sustainability, University of Iowa, Iowa City, IA, USA

Dates: May 2023 – Present

- Built repeatable Python + PostgreSQL/PostGIS pipelines to clean, standardize, and geocode 280M historical place records into analysis-ready county/city outputs with validation summaries.
- Produced high-quality maps, diagrams, and visualizations of mobility patterns and spatial networks for peer-reviewed journals, proposals, and presentations.
- Integrated and validated multi-source spatial reference datasets and modified spatial algorithms to produce defensible fine-resolution geocoding results.
- Developed GIS-based modeling workflows to extract and analyze location changes for 30M+ individuals across multiple decades.
- Improved a 60M-node spatial network via rule-based deduplication and data imputation, recovering 10M connections for downstream spatial modeling.

Graduate Research Associate

Hydroscience & Engineering Department, University of Iowa, Iowa City, IA, USA

Dates: May 2024 – August 2024

- Built the “[Rivers and Risks](#)” StoryMap using ArcGIS Online, and Survey123 to support outreach and stakeholder communication.
- Translated engineer analysis of water quality risk assessments to maps, visualizations, and stories to communication to broad audiences.
- Analyzed water pollution and social vulnerability indicators linkage to assess exposure risk patterns.
- Created an animated map of Iowa population growth and transportation development using JavaScript and D3.js.

Teaching Assistant

Department of Geographical and Sustainability Sciences, University of Iowa, Iowa City, IA, USA

Dates: August 2022 – May 2023

- Taught two hands-on GIS laboratory sections and mentored student projects in ArcGIS Pro, spatial analysis workflows, and cartographic outputs.
- Mentored 42+ students by debugging and troubleshooting GIS project issues involving vector and raster data.

Spatial Data Analyst

RassadIran Surveying Company, Tehran, Iran

Dates: April 2014 – March 2015

- Evaluated elevation and surveying errors in road construction projects and communicated results to engineering teams for corrective action.
- Supported cadastral map development for Tehran in ArcGIS by digitizing parcel boundaries and base layers.

EDUCATION

Ph.D., Geography - Geographic Information Science and Cartography

University of Iowa, Iowa City, IA, USA

Dates: August 2022 – December 2026 (Expected)

M.Sc., Remote Sensing and Geographic Information Systems - Satellite Meteorology

Science and Research Branch, Islamic Azad University, Tehran, Iran

Dates: September 2017 – September 2020

B.E., Surveying Engineering

Shahid Rajaei Teacher Training University, Tehran, Iran

Dates: February 2012 – March 2014

TECHNICAL SKILLS

- **Programming Language:** Python (Arcpy, Geopandas, Scikit-learn, NetworkX, Matplotlib), R (tidyverse, sf, ggplot2), SQL (PostgreSQL/PostGIS), JavaScript (D3.js).
- **Software:** QGIS, ArcGIS (Pro, Online, StoryMaps, Survey123), pgAdmin.
- **Dev Tools:** Linux, Jupyter, Git/GitHub, PyCharm, VS Code.
- **Methods:** Spatial analysis and visualization, Raster and vector analysis, Cartography, GIS-based modeling, Workflow documentation.

SELECTED PROJECTS (REMOTE SENSING & TERRAIN)

- **Landsat Image Processing:** Processed and classified multi-spectral satellite imagery to derive land-surface indicators and produced map layouts. (Fall 2018)
- **Terrain Modeling:** Built DEM-derived terrain surface layers for modeling groundwater risk assessment (2018).
- **GIS-based Hydrological Analysis:** Developed and compares several new GIS-based models for groundwater vulnerability estimation in the largest aquifer in Iran (2019-2020).

AWARDS AND HONORS

- Graduate College Post-Comprehensive Research Fellowship, University of Iowa (\$11,375), Spring 2026
- Digital Scholarship & Publishing Summer Fellowship, University of Iowa (\$5500), Summer 2025
- Best Student Paper Award, CaGIS + UCGIS Symposium (\$150), June 2024
- Regional Cartographic Conference Scholarship (\$200), June 2024

CERTIFICATIONS (SELECTED)

- Esri Spatial Data Science: The New Frontier in Analytics, Esri, November 2025
- Interdisciplinary Graduate Certificate in Informatics, University of Iowa, May 2025
- Artificial Intelligence Proficiency Level 1–3, The Iowa Initiative for Artificial Intelligence, March 2024 – May 2025

PUBLICATIONS (1 of 9)

- **Torkashvand, M.,** Neshat, A., Javadi, S., Pradhan, B. (2021) New hybrid evolutionary algorithm for optimizing index-based groundwater vulnerability assessment method, [Journal of Hydrology](#), 598, 126446.

LEADERSHIP AND SERVICE

- GIS Lab Representative, Geography Graduate Student Organization, University of Iowa, August 2024 – May 2025
- Guest Lecturer, GEOG 2050: Foundations of GIS, University of Iowa, Fall 2023